

Physics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- Draw diagrams using pencil
- Board-approved calculators may be used
- A data sheet, formulae sheets and Periodic Table are provided at the back of this paper

Total marks – 100

Section I Pages 2–28

75 marks

This section has two parts, Part A and Part B

Part A – 20 marks

- Attempt Questions 1–20
- Allow about 35 minutes for this part

Part B – 55 marks

- Attempt Questions 21–30
- Allow about 1 hour and 40 minutes for this part

Section II Pages 29–37

25 marks

- Attempt ONE question from Questions 31–35
- Allow about 45 minutes for this section

Section I

75 marks

Part A – 20 marks

Attempt Questions 1–20

Allow about 35 minutes for this part

Use the multiple-choice answer sheet for Questions 1–20.

- 1** Some mobile phones are recharged at a power point using a charger that contains a transformer.

What is the purpose of the transformer?

- (A) To convert AC at the power point to DC
- (B) To convert DC at the power point to AC
- (C) To increase the AC voltage at the power point
- (D) To decrease the AC voltage at the power point

- 2** Both Westinghouse and Edison proposed methods for supplying electricity to homes.

What advantage did Westinghouse's method have over Edison's?

- (A) DC could be supplied at a lower current, making it safer for the consumer.
- (B) DC could be supplied at a lower voltage, making it safer for the consumer.
- (C) AC allowed for the use of a lower transmission current, reducing energy losses.
- (D) AC allowed for the use of a lower transmission voltage, reducing energy losses.

- 3** A region of space contains a constant magnetic field and a constant electric field.

How will these fields affect an electron that is stationary in this region?

- (A) Both fields will exert a force.
- (B) Neither field will exert a force.
- (C) Only the electric field will exert a force.
- (D) Only the magnetic field will exert a force.

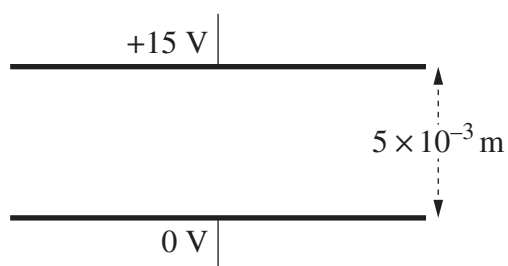
- 4 The graph shows information about three superconductors *X*, *Y* and *Z* including their year of discovery.



Which row of the table correctly classifies *X*, *Y* and *Z*?

	<i>X</i>	<i>Y</i>	<i>Z</i>
(A)	Metal	Metal alloy	Compound
(B)	Metal	Compound	Metal alloy
(C)	Metal alloy	Metal	Compound
(D)	Metal alloy	Compound	Metal

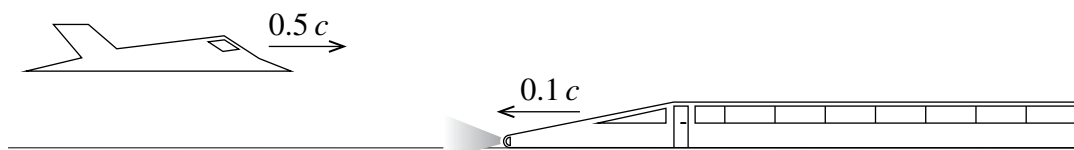
- 5 The diagram shows two parallel charged plates 5×10^{-3} m apart.



What is the magnitude of the electric field between the plates in V m^{-1} ?

- (A) 3.3×10^{-4}
 (B) 0.33
 (C) 3
 (D) 3000

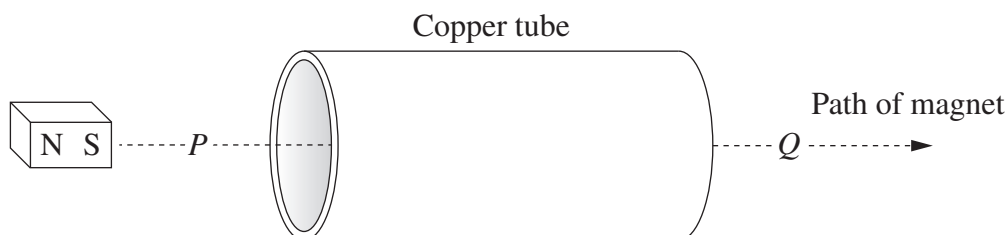
- 6 In a thought experiment, a jet is travelling at $0.5c$ relative to the ground, towards a train that is travelling at $0.1c$ relative to the ground, as shown.



What is the speed of the light emitted from the train's headlight, as measured by a pilot in the jet?

- (A) $0.1c$
 (B) $0.4c$
 (C) $0.6c$
 (D) $1.0c$
- 7 A magnet passes through a copper tube at constant velocity along the path shown.

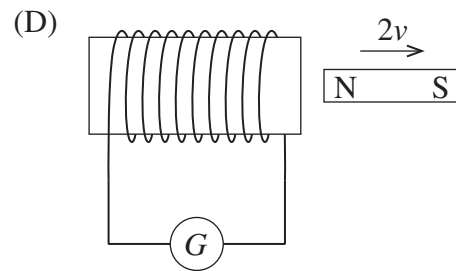
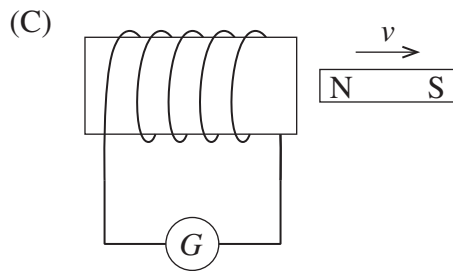
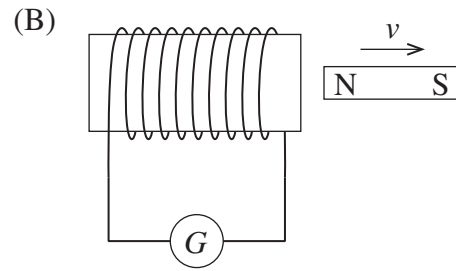
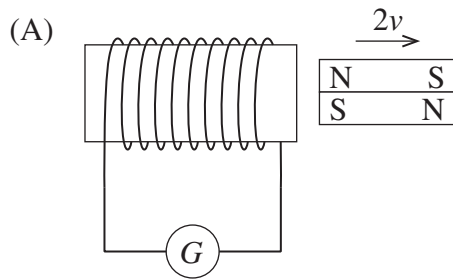
A current is induced in the tube by the motion of the magnet.



Which row of the table correctly describes the forces acting between the tube and the magnet at points P and Q ?

	<i>Force at P</i>	<i>Force at Q</i>
(A)	Attraction	Repulsion
(B)	Repulsion	Attraction
(C)	Attraction	Attraction
(D)	Repulsion	Repulsion

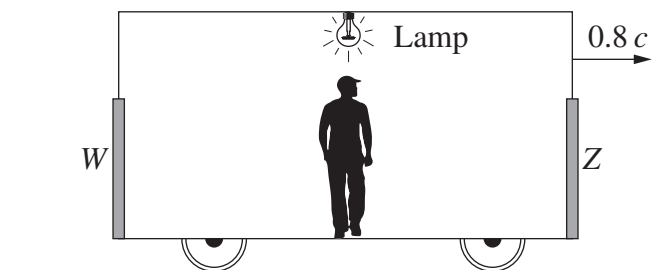
- 8 Which movement of the magnet(s) will produce the greatest deflection of the galvanometer?



- 9 How does back emf affect a DC motor?

- (A) It creates heat in the iron core.
- (B) It limits the speed of the motor.
- (C) It reverses the current in the coil.
- (D) It increases the torque of the motor.

- 10 In a thought experiment, a train is moving at a constant speed of $0.8c$. A lamp is located at the midpoint of a carriage. There are doors W and Z at each end of the carriage which open automatically when light from the lamp reaches them.

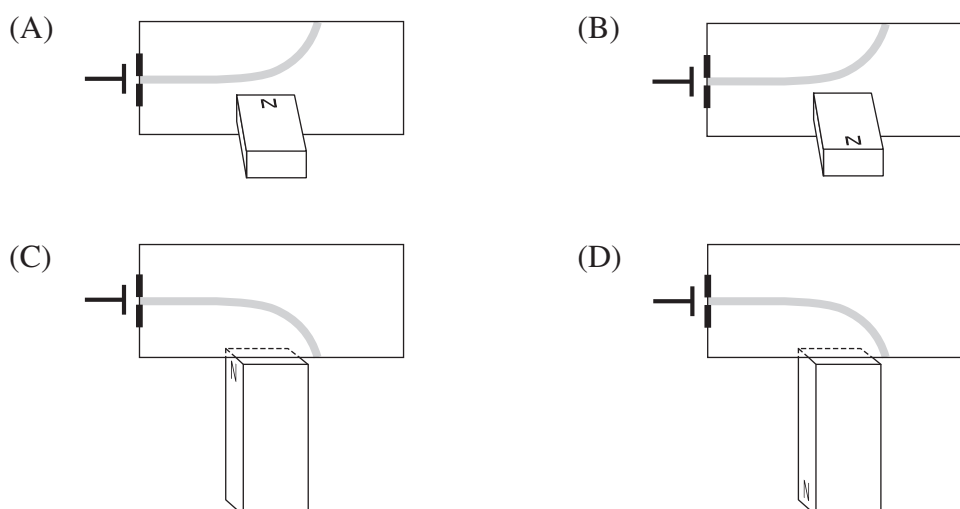


The passenger standing at the midpoint of the carriage switches on the lamp.

Which statement best explains what the passenger observes about the doors?

- (A) Z opens before W because the lamp is moving towards Z .
 - (B) W opens before Z because W is moving towards the lamp.
 - (C) W and Z open simultaneously because the lamp is placed at an equal distance from both.
 - (D) W and Z open simultaneously because the distance from the lamp to each door has contracted by the same amount.
- 11 What is the wavelength, in metres, of a photon with an energy of 3.5 eV ?
- (A) 1.2×10^{-6}
 - (B) 3.5×10^{-7}
 - (C) 1.18×10^{-15}
 - (D) 5.67×10^{-26}

12 Which diagram correctly shows the deflection of a cathode ray by a bar magnet?

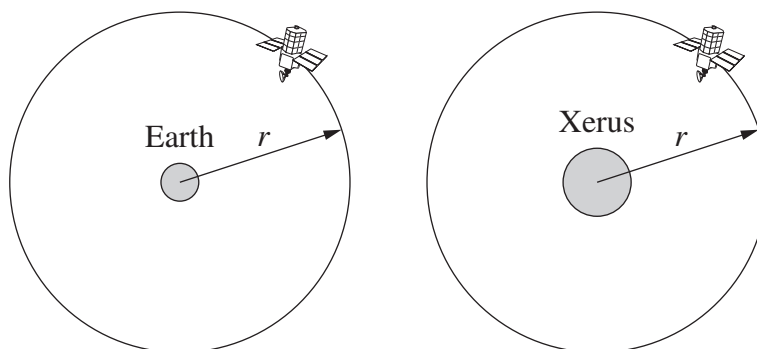


13 When light of a specific frequency strikes a metal surface, photoelectrons are emitted.

If the light intensity is increased but the frequency remains the same, which row of the table is correct?

	<i>Number of photoelectrons emitted</i>	<i>Maximum kinetic energy of the photoelectrons</i>
(A)	Remains the same	Remains the same
(B)	Remains the same	Increases
(C)	Increases	Remains the same
(D)	Increases	Increases

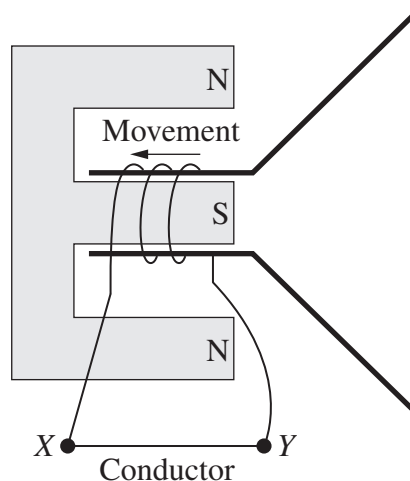
- 14 A satellite orbits Earth with period T . An identical satellite orbits the planet Xerus which has a mass four times that of Earth. Both satellites have the same orbital radius r .



What is the period of the satellite orbiting Xerus?

- (A) $\frac{T}{4}$
 - (B) $\frac{T}{2}$
 - (C) $2T$
 - (D) $4T$
- 15 Which of the following statements correctly describes the effect of light in a solar cell?
- (A) The light produces electron-hole pairs.
 - (B) The light causes pure silicon to become n -type silicon.
 - (C) The light increases the potential difference across the p - n boundary.
 - (D) The light transfers energy to electrons in the conduction band of the p -type semiconductor.

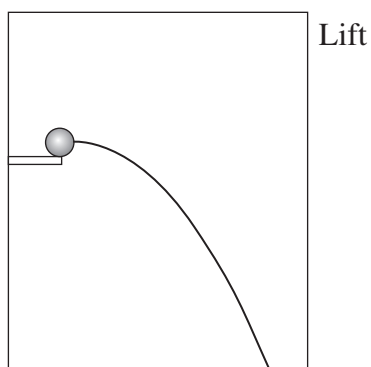
- 16 The cone of a speaker is pushed so that the coil moves in the direction shown.



Which row of the table correctly identifies the behaviour of the speaker and the direction of the current through the conductor?

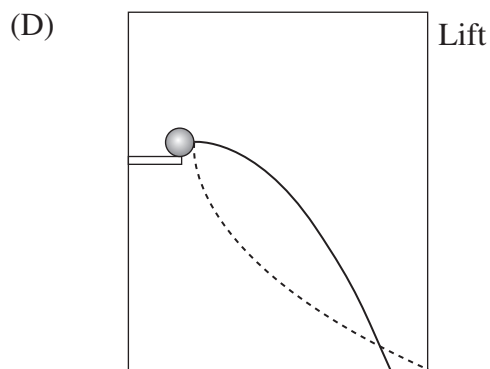
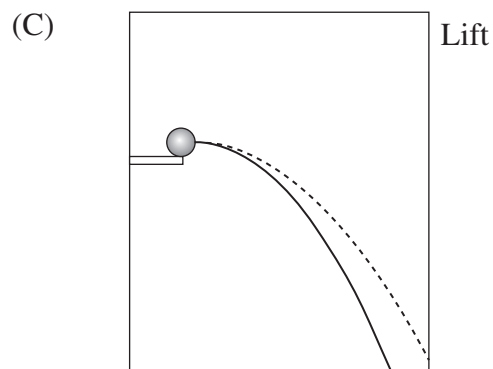
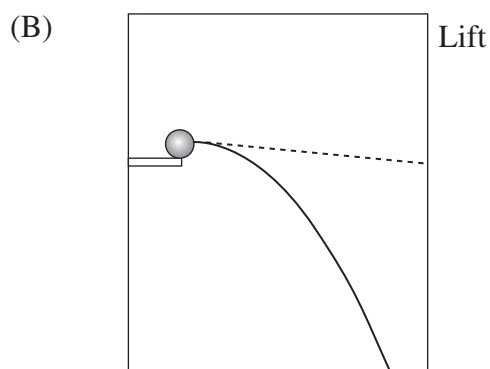
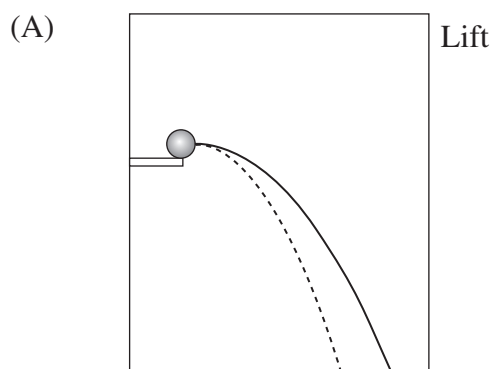
	<i>The speaker behaves like a ...</i>	<i>The direction of the current is from ...</i>
(A)	generator	<i>X to Y</i>
(B)	generator	<i>Y to X</i>
(C)	motor	<i>X to Y</i>
(D)	motor	<i>Y to X</i>

- 17 A projectile was launched horizontally inside a lift in a building. The diagram shows the path of the projectile when the lift was stationary.

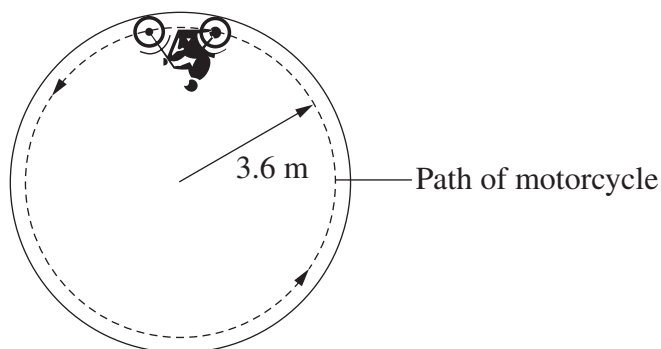


The projectile was launched again with the same velocity. At this time, the lift was slowing down as it approached the top floor of the building.

Which diagram correctly shows the new path of the projectile (dotted line) relative to the path created in the stationary lift (solid line)?



- 18 A motorcycle travels around a vertical circular path of radius 3.6 m at a constant speed. The combined mass of the rider and motorcycle is 200 kg.



What is the minimum speed, in m s^{-1} , at which the motorcycle must travel to maintain the circular path?

- (A) 0.42
 (B) 1.9
 (C) 5.9
 (D) 35
- 19 Muons are subatomic particles which at rest have a lifetime of 2.2 microseconds (μs). When they are produced in Earth's upper atmosphere, they travel at $0.9999c$.

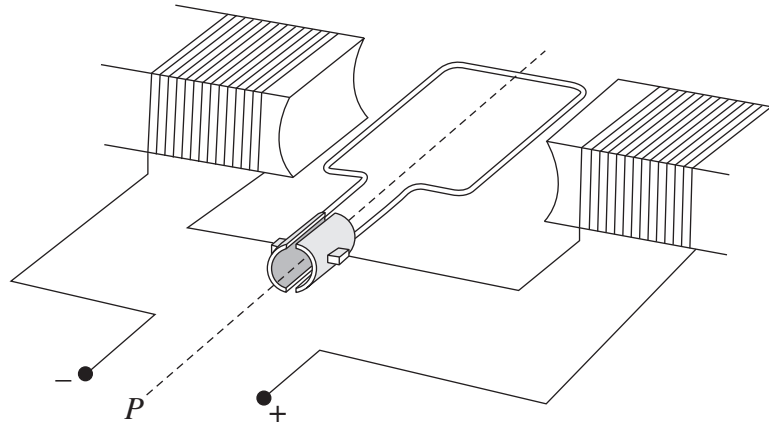
Using classical physics, the distance travelled by a muon in its lifetime can be calculated as follows:

$$x = vt \\ = 660 \text{ m}$$

Which row of the table correctly summarises the behaviour of these muons?

Muon's reference frame		Earth's reference frame	
<i>Distance travelled</i> (m)	<i>Lifetime</i> (μs)	<i>Distance travelled</i> (m)	<i>Lifetime</i> (μs)
(A) 660	2.2	> 660	> 2.2
(B) > 660	> 2.2	660	2.2
(C) 660	2.2	< 660	< 2.2
(D) < 660	< 2.2	660	2.2

- 20 In the motor shown, the rotor spins clockwise, as viewed from point P , when connected to a DC supply.



What happens when the motor is connected to an AC supply?

- (A) There is no movement of the rotor.
- (B) The rotor produces clockwise movement only.
- (C) The rotor vibrates at the frequency of the AC supply.
- (D) The rotor continuously turns half a rotation clockwise, then half a rotation anticlockwise.

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Centre Number

Section I (continued)

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Student Number

Part B – 55 marks**Attempt Questions 21–30****Allow about 1 hour and 40 minutes
for this part**

Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.

Show all relevant working in questions involving calculations.

Extra writing space is provided on pages 27 and 28. If you use this space, clearly indicate which question you are answering.

Write your Centre Number and Student Number at the top of this page.

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Question 21 (5 marks)

- (a) Why does orbital decay occur more rapidly for satellites in a low-Earth orbit than for satellites in other orbits? **2**

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- (b) Calculate the magnitude of the gravitational force that acts on a 50 kg satellite when it is 8000 km from Earth’s centre. **3**

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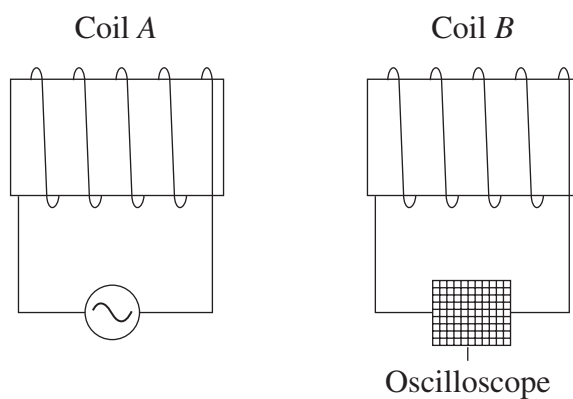
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Question 22 (6 marks)

- (a) When an alternating current is passed through coil *A*, a voltage is observed on the oscilloscope connected to coil *B*.

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How could a bar magnet be used, instead of coil *A*, to produce a similar pattern on the oscilloscope?

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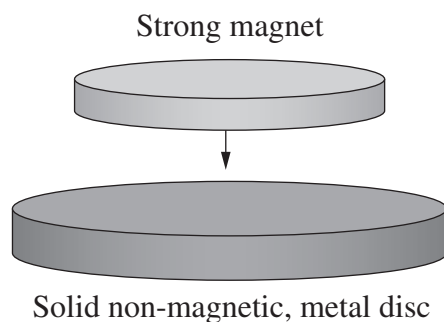
Question 22 continues on page 17

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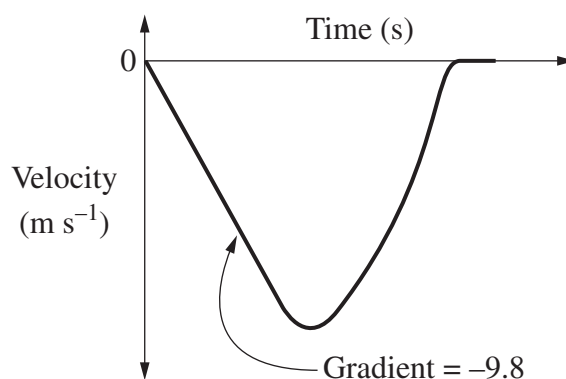
Question 22 (continued)

- (b) A strong magnet is at rest a few centimetres above a solid metal disc made of a non-magnetic metal. The magnet is then dropped.

4



The velocity of the magnet is shown in this graph.



Account for the shape of the graph.

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End of Question 22

Question 23 (6 marks)

- (a) Explain the role of the deflection plates in displaying voltage changes applied to the input of a cathode ray oscilloscope.

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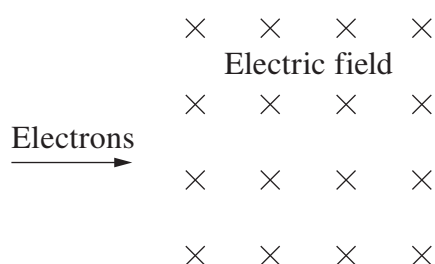
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- (b) The diagram shows electrons travelling in a vacuum at $5.2 \times 10^4 \text{ m s}^{-1}$ entering an electric field of 10 V m^{-1} .

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A magnetic field is applied so that the electrons continue undeflected.

What is the magnitude and direction of the magnetic field?

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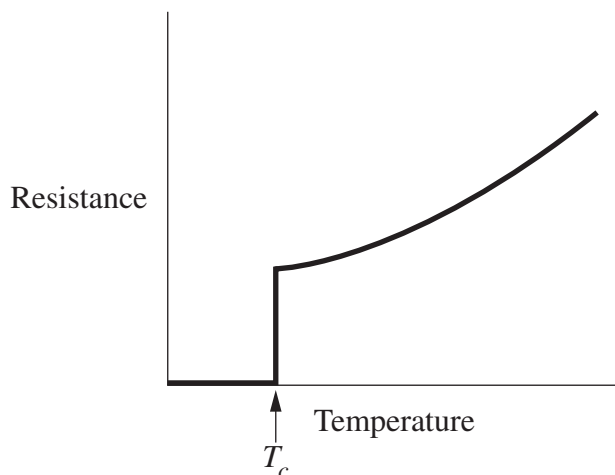
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Question 24 (4 marks)

The graph shows the effect of temperature on the resistance of a metal wire.

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Explain the shape of this graph in terms of the behaviour of electrons in the metal, both above and below the critical temperature (T_c).

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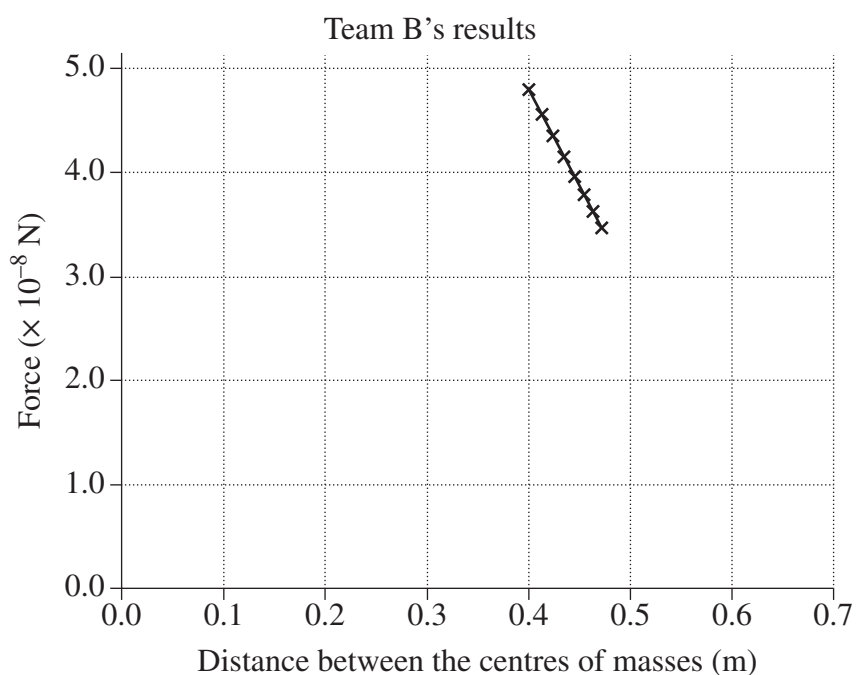
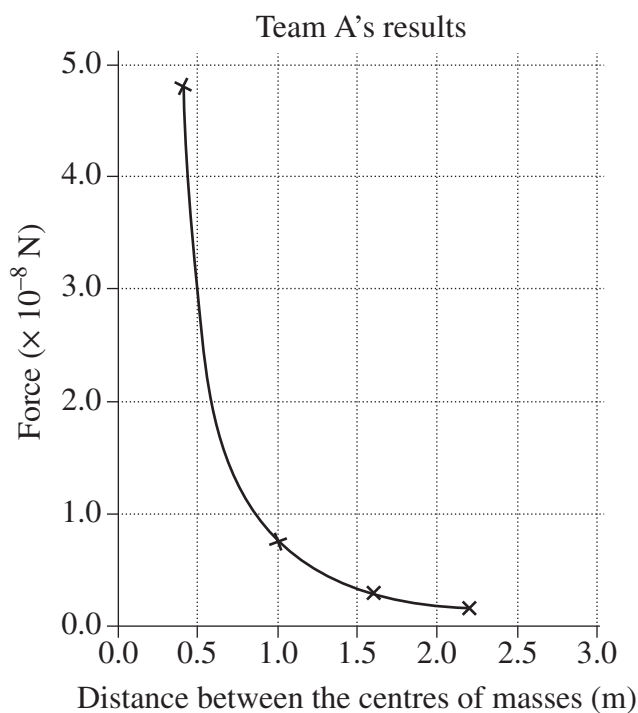
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Question 25 (5 marks)

Two teams carried out independent experiments with the purpose of investigating Newton's Law of Universal Gravitation. Each team used the same procedure to accurately measure the gravitational force acting between two spherical masses over a range of distances.

The following graphs show the data collected by each team.



Question 25 continues on page 21

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Question 25 (continued)

- (a) Compare qualitatively the relationship between force and distance in the graphs. **2**

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- (b) Assess the appropriateness of Team A's data and Team B's data in achieving the purpose of the experiments. **3**

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End of Question 25

Question 26 (5 marks)

Explain how applications of transformers have affected society.

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Question 27 (7 marks)

- (a) Outline observations that Hertz made that were consistent with the photoelectric effect. In your answer, identify equipment that was essential for him to make these observations. **3**

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- (b) Explain how the result of ONE investigation of the photoelectric effect changed the scientific understanding of the nature of light. **4**

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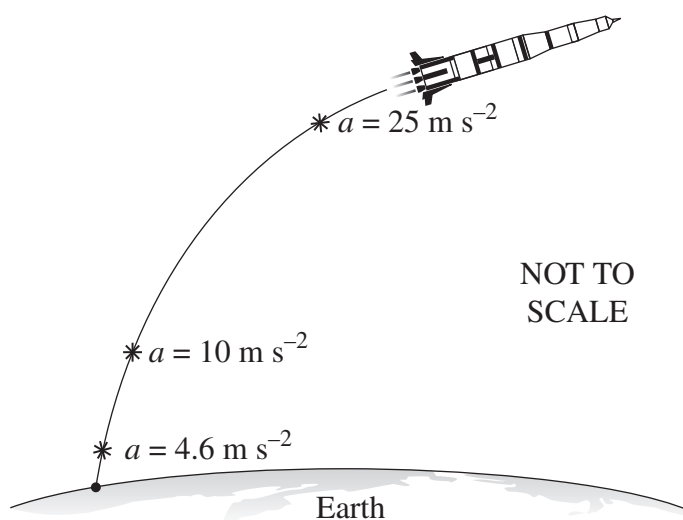
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Question 28 (5 marks)

The following diagram shows the acceleration of a rocket during the first stage of its launch.

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Explain the acceleration of the rocket with reference to the law of conservation of momentum.

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Question 29 (6 marks)

Explain how different discoveries in physics led to the development of THREE technologies, including the electric generator.

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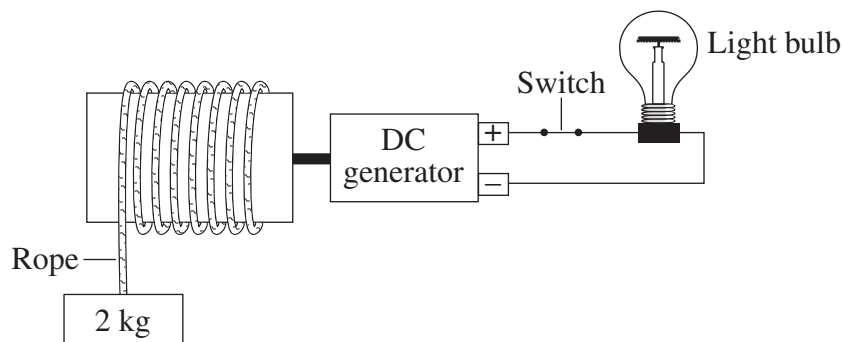
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Question 30 (6 marks)

The following makeshift device was made to provide lighting for a stranded astronaut on Mars.

The mass of Mars is 6.39×10^{23} kg.



The 2 kg mass falls, turning the DC generator, which supplies energy to the light bulb. The mass falls from a point that is 3 376 204 m from the centre of Mars.

- (a) Calculate the maximum possible energy released by the light bulb as the mass falls through a distance of one metre. 3

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- (b) Explain the difference in the behaviour of the falling mass when the switch is open. 3

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Section I Part B extra writing space

If you use this space, clearly indicate which question you are answering.

Handwriting practice area with horizontal lines.

Section I Part B extra writing space

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Physics

Section II

25 marks

Attempt ONE question from Questions 31–35

Allow about 45 minutes for this section

Answer parts (a)–(e) of one question in the Section II Writing Booklet. Extra writing booklets are available.

Show all relevant working in questions involving calculations.

	Pages
Question 31 Geophysics	30–31
Question 32 Medical Physics	32
Question 33 Astrophysics	33
Question 34 From Quanta to Quarks	34
Question 35 The Age of Silicon	35–37

Question 31 — Geophysics (25 marks)

Answer parts (a), (b) and (c) of the question on pages 2–4 of the Section II Writing Booklet. Start each part of the question on a new page.

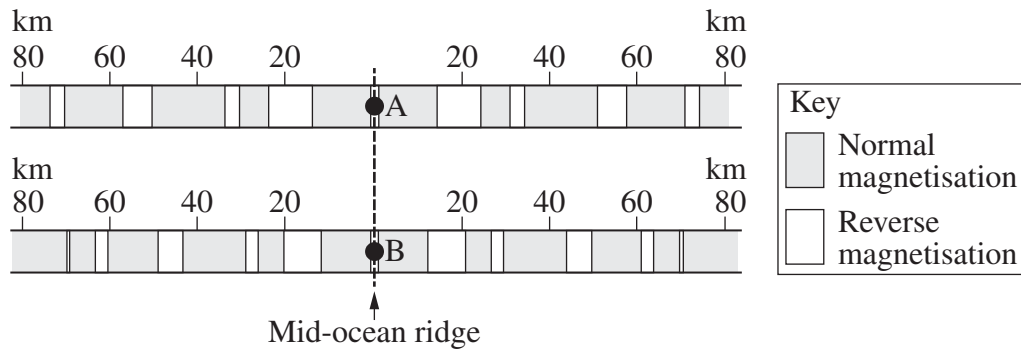
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|-----|------|--|----------|
| (a) | (i) | What role do geophysicists play in the monitoring of the nuclear test ban treaties? | 2 |
| | (ii) | Explain how ONE geophysical method used in mineral exploration can produce an economic benefit. | 3 |
| (b) | (i) | Describe a model that shows the relationship between the inclination of Earth's magnetic field and latitude. | 3 |
| | (ii) | Describe features of Earth's magnetic field, other than the variation of its inclination with latitude. | 3 |
| (c) | | Explain TWO corrections that must be made when gravimeter data is being analysed. | 4 |

Question 31 continues on page 31

Question 31 (continued)

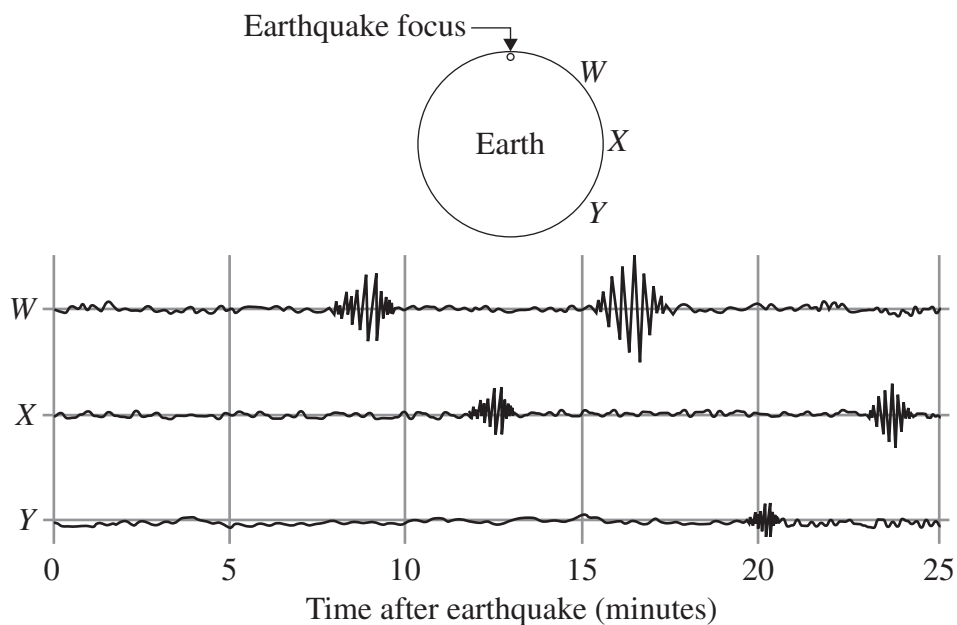
Answer parts (d) and (e) of the question on pages 6–8 of the Section II Writing Booklet. Start each part of the question on a new page.

- (d) The diagram shows the magnetisation of rocks along lines perpendicular to the same mid-ocean ridge, from locations A and B. 4



What valid inferences can be made from these data sets? Justify your answer.

- (e) After an earthquake, the following seismograms were obtained from stations W, X and Y on Earth's surface. 6



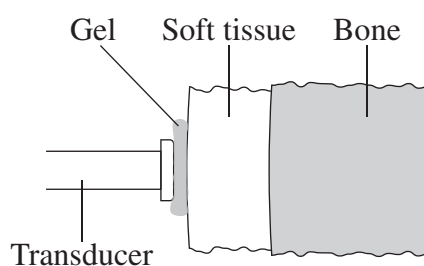
Relate features on the seismograms to the behaviour of different types of earthquake waves and Earth's internal structure.

End of Question 31

Question 32 — Medical Physics (25 marks)

Answer parts (a), (b) and (c) of the question on pages 2–4 of the Section II Writing Booklet. Start each part of the question on a new page.

- (a) (i) What are the differences between hard and soft X-rays? 2
- (ii) Account for the widespread use of X-rays to create images of human limbs. 3
- (b) (i) Describe the piezoelectric effect and its application in ultrasound imaging. 3
- (ii) An ultrasound pulse leaves the transducer and passes from the gel through soft tissue and bone as shown. A table of acoustic impedances is provided. 3



<i>Medium</i>	<i>Acoustic impedance (kg m⁻² s⁻¹)</i>
Gel	1.632×10^6
Soft tissue	1.632×10^6
Bone	6.732×10^6

Describe what happens to the intensity of the ultrasound pulse at each boundary. Use calculations to support your answer.

- (c) Contrast the use of Doppler ultrasound imaging with the use of computed axial tomography (CAT) imaging. In your answer, include an example of how each is used. 4

Answer parts (d) and (e) of the question on pages 6–8 of the Section II Writing Booklet. Start each part of the question on a new page.

- (d) Describe how positron emission tomography (PET) scanning is used in diagnosis. In your answer, refer to the interaction of positrons and electrons. 4
- (e) Explain how principles of physics are applied in magnetic resonance imaging (MRI). 6

Question 33 — Astrophysics (25 marks)

Answer parts (a), (b) and (c) of the question on pages 2–4 of the Section II Writing Booklet. Start each part of the question on a new page.

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|-----|------|--|----------|
| (a) | (i) | Distinguish between the terms <i>resolution</i> and <i>sensitivity</i> as applied to optical telescopes. | 2 |
| | (ii) | Outline ways of improving the resolution of ground-based telescopes. | 3 |
| (b) | (i) | Explain how trigonometric parallax is used to determine the distance to a star. | 3 |
| | (ii) | Determine the absolute magnitude of a star which has an apparent magnitude of 4.2 and a parallax of 0.055 arc seconds. | 3 |
| (c) | | Describe how the spectrum of a star can be used to determine its chemical composition and surface temperature. | 4 |

Answer parts (d) and (e) of the question on pages 6–8 of the Section II Writing Booklet. Start each part of the question on a new page.

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|-----|--|---|----------|
| (d) | | Explain how the study of different types of variable stars has increased our understanding of celestial objects. | 4 |
| (e) | | Describe how the distribution of stars on a Hertzsprung–Russell diagram relates to the processes that occur during their evolution. | 6 |

Question 34 — From Quanta to Quarks (25 marks)

Answer parts (a), (b) and (c) of the question on pages 2–4 of the Section II Writing Booklet. Start each part of the question on a new page.

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|-----|------|--|----------|
| (a) | (i) | Identify the TWO types of nucleon and state ONE difference between them. | 2 |
| | (ii) | Explain the stability of ${}^4_2\text{He}$ nuclei in terms of TWO forces. | 3 |
| (b) | (i) | Outline features of the hydrogen spectrum that Bohr's model could not explain. | 3 |
| | (ii) | Calculate the initial energy level of an electron in a hydrogen atom if it emitted $4.089 \times 10^{-19} \text{ J}$ on transition to the $n = 2$ level. | 3 |
| (c) | | Describe how contributions to atomic theory made by Pauli and Heisenberg changed Bohr's model of the atom. | 4 |

Answer parts (d) and (e) of the question on pages 6–8 of the Section II Writing Booklet. Start each part of the question on a new page.

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|-----|--|--|----------|
| (d) | | Explain how evidence from experiments involving particle accelerators and detectors has provided support for the standard model of matter. | 4 |
| (e) | | How did Fermi's work in nuclear physics lead to applications that have affected society? | 6 |

Question 35 — The Age of Silicon (25 marks)

Answer parts (a), (b) and (c) of the question on pages 2–4 of the Section II Writing Booklet. Start each part of the question on a new page.

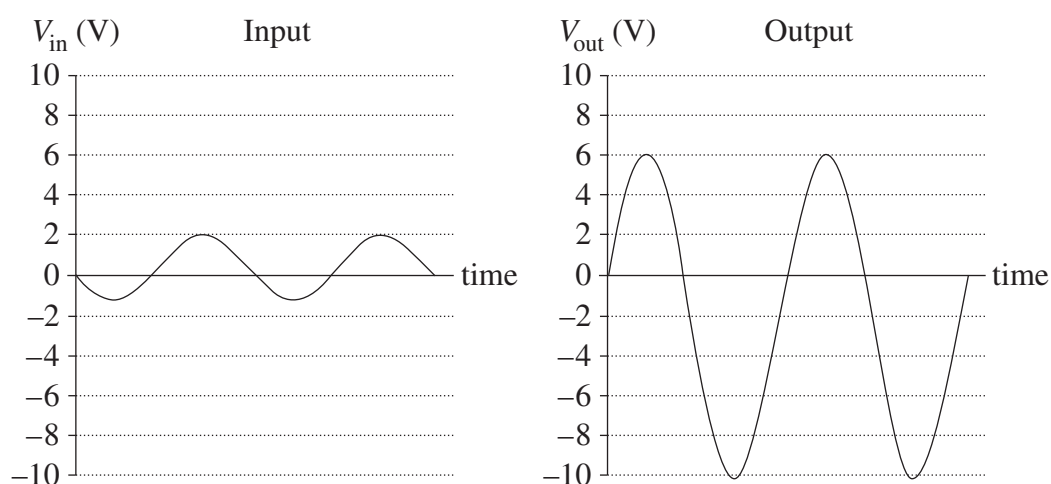
- (a) (i) Use sketches to show how analogue and digital voltage outputs are different. **2**
- (ii) Describe ONE advantage of electronic circuits and ONE advantage of electric circuits. **3**

- (b) (i) The table shows properties of a real operational amplifier. **3**

Voltage gain	200 000
Input impedance	2 M Ω
Output resistance	100 Ω
Bandwidth	0–1 MHz

How do these properties compare with those of an ideal amplifier?

- (ii) An inverting amplifier is constructed using a single operational amplifier and two resistors. The input resistor has a value of 2.2 k Ω . The graphs show input and output voltages for the circuit of this amplifier over the same time interval. **3**



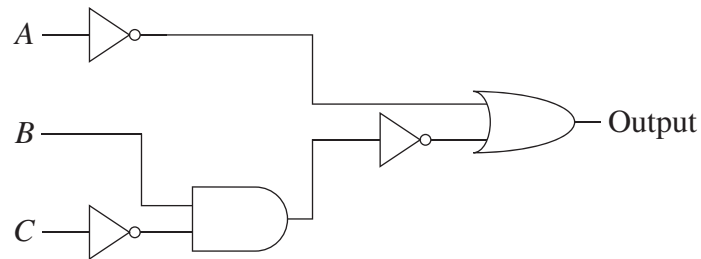
Draw a diagram of this circuit, showing the value of each resistor.

Question 35 continues on page 36

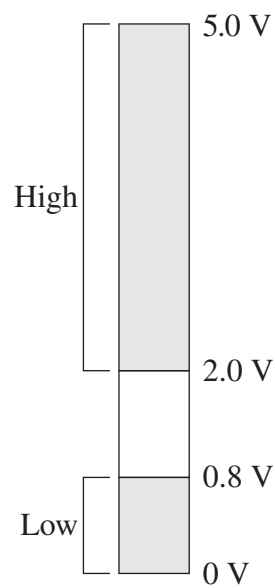
Question 35 (continued)

- (c) The logic circuit shown was constructed using a number of components.

4



The diagram shows acceptable input voltages for these components.



A low voltage output is required from the circuit.

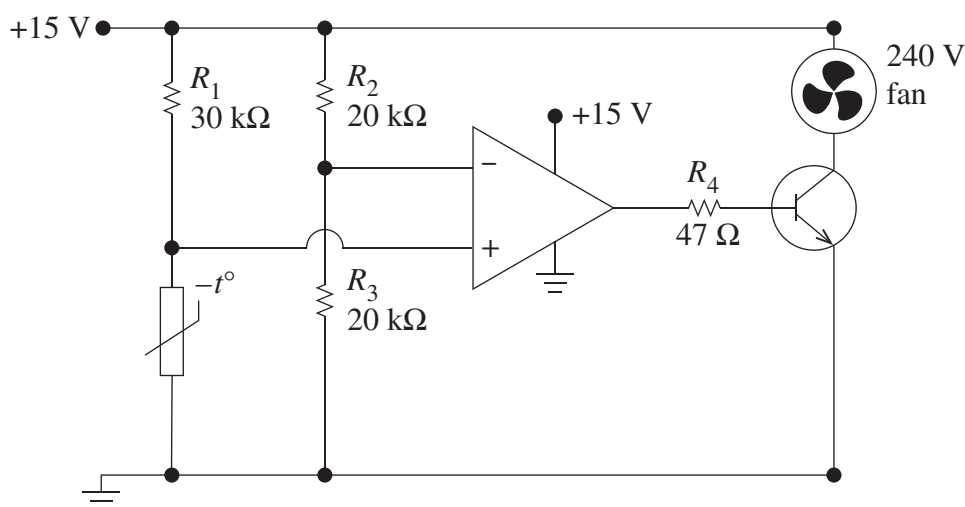
Determine possible input voltages for *A*, *B* and *C*. Support your answer using a truth table.

Question 35 continues on page 37

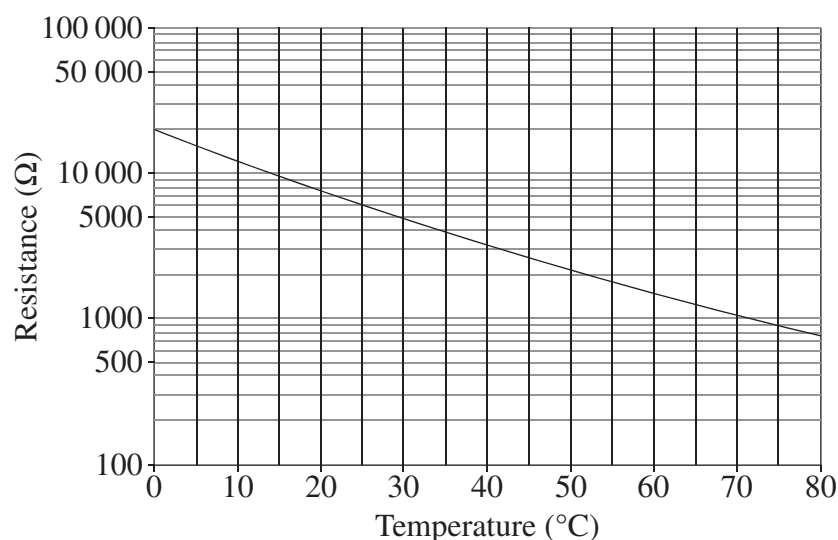
Question 35 (continued)

Answer parts (d) and (e) of the question on pages 6–8 of the Section II Writing Booklet. Start each part of the question on a new page.

- (d) Explain the advantages of using integrated circuits instead of individual transistors in electronic devices. 4
- (e) A circuit is required to automatically switch on a 240 V fan when the temperature exceeds 30°C. The circuit shown was designed by a student but does not operate as intended. 6



The graph shows the characteristic curve of the thermistor in the circuit.



Explain THREE changes that need to be made for this circuit to operate correctly.

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DATA SHEET

Charge on electron, q_e	$-1.602 \times 10^{-19} \text{ C}$
Mass of electron, m_e	$9.109 \times 10^{-31} \text{ kg}$
Mass of neutron, m_n	$1.675 \times 10^{-27} \text{ kg}$
Mass of proton, m_p	$1.673 \times 10^{-27} \text{ kg}$
Speed of sound in air	340 m s^{-1}
Earth's gravitational acceleration, g	9.8 m s^{-2}
Speed of light, c	$3.00 \times 10^8 \text{ m s}^{-1}$
Magnetic force constant, $\left(k \equiv \frac{\mu_0}{2\pi}\right)$	$2.0 \times 10^{-7} \text{ N A}^{-2}$
Universal gravitational constant, G	$6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
Mass of Earth	$6.0 \times 10^{24} \text{ kg}$
Planck constant, h	$6.626 \times 10^{-34} \text{ J s}$
Rydberg constant, R (hydrogen)	$1.097 \times 10^7 \text{ m}^{-1}$
Atomic mass unit, u	$1.661 \times 10^{-27} \text{ kg}$ $931.5 \text{ MeV}/c^2$
1 eV	$1.602 \times 10^{-19} \text{ J}$
Density of water, ρ	$1.00 \times 10^3 \text{ kg m}^{-3}$
Specific heat capacity of water	$4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

FORMULAE SHEET

$$v = f\lambda$$

$$I \propto \frac{1}{d^2}$$

$$\frac{v_1}{v_2} = \frac{\sin i}{\sin r}$$

$$E = \frac{F}{q}$$

$$R = \frac{V}{I}$$

$$P = VI$$

$$\text{Energy} = VIt$$

$$v_{\text{av}} = \frac{\Delta r}{\Delta t}$$

$$a_{\text{av}} = \frac{\Delta v}{\Delta t} \quad \text{therefore} \quad a_{\text{av}} = \frac{v - u}{t}$$

$$\Sigma F = ma$$

$$F = \frac{mv^2}{r}$$

$$E_k = \frac{1}{2}mv^2$$

$$W = Fs$$

$$p = mv$$

$$\text{Impulse} = Ft$$

$$E_p = -G \frac{m_1 m_2}{r}$$

$$F = mg$$

$$v_x^2 = u_x^2$$

$$v = u + at$$

$$v_y^2 = u_y^2 + 2a_y \Delta y$$

$$\Delta x = u_x t$$

$$\Delta y = u_y t + \frac{1}{2}a_y t^2$$

$$\frac{r^3}{T^2} = \frac{GM}{4\pi^2}$$

$$F = \frac{Gm_1 m_2}{d^2}$$

$$E = mc^2$$

$$l_v = l_0 \sqrt{1 - \frac{v^2}{c^2}}$$

$$t_v = \frac{t_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$m_v = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

FORMULAE SHEET

$$\frac{F}{l} = k \frac{I_1 I_2}{d}$$

$$d = \frac{1}{p}$$

$$F = BIl \sin \theta$$

$$M = m - 5 \log \left(\frac{d}{10} \right)$$

$$\tau = Fd$$

$$\frac{I_A}{I_B} = 100^{(m_B - m_A)/5}$$

$$\tau = nBIA \cos \theta$$

$$m_1 + m_2 = \frac{4\pi^2 r^3}{GT^2}$$

$$\frac{V_p}{V_s} = \frac{n_p}{n_s}$$

$$F = qvB \sin \theta$$

$$\frac{1}{\lambda} = R \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$

$$E = \frac{V}{d}$$

$$\lambda = \frac{h}{mv}$$

$$E = hf$$

$$c = f\lambda$$

$$A_0 = \frac{V_{\text{out}}}{V_{\text{in}}}$$

$$Z = \rho v$$

$$\frac{V_{\text{out}}}{V_{\text{in}}} = -\frac{R_{\text{f}}}{R_{\text{i}}}$$

$$\frac{I_r}{I_0} = \frac{[Z_2 - Z_1]^2}{[Z_2 + Z_1]^2}$$

PERIODIC TABLE OF THE ELEMENTS

1 H 1.008 Hydrogen		KEY										2 He 4.003 Helium								
3 Li 6.941 Lithium	4 Be 9.012 Beryllium		Atomic Number Symbol		79 Au 197.0 Gold		Standard Atomic Weight Name		5 B 10.81 Boron		6 C 12.01 Carbon		7 N 14.01 Nitrogen		8 O 16.00 Oxygen		9 F 19.00 Fluorine		10 Ne 20.18 Neon	
	11 Na 22.99 Sodium		12 Mg 24.31 Magnesium						13 Al 26.98 Aluminium		14 Si 28.09 Silicon		15 P 30.97 Phosphorus		16 S 32.07 Sulfur		17 Cl 35.45 Chlorine		18 Ar 39.95 Argon	
19 K 39.10 Potassium	20 Ca 40.08 Calcium	21 Sc 44.96 Scandium	22 Ti 47.87 Titanium	23 V 50.94 Vanadium	24 Cr 52.00 Chromium	25 Mn 54.94 Manganese	26 Fe 55.85 Iron	27 Co 58.93 Cobalt	28 Ni 58.69 Nickel	29 Cu 63.55 Copper	30 Zn 65.38 Zinc	31 Ga 69.72 Gallium	32 Ge 72.64 Germanium	33 As 74.92 Arsenic	34 Se 78.96 Selenium	35 Br 79.90 Bromine	36 Kr 83.80 Krypton			
37 Rb 85.47 Rubidium	38 Sr 87.61 Strontium	39 Y 88.91 Yttrium	40 Zr 91.22 Zirconium	41 Nb 92.91 Niobium	42 Mo 95.96 Molybdenum	43 Tc Technetium	44 Ru 101.1 Ruthenium	45 Rh 102.9 Rhodium	46 Pd 106.4 Palladium	47 Ag 107.9 Silver	48 Cd 112.4 Cadmium	49 In 114.8 Indium	50 Sn 118.7 Tin	51 Sb 121.8 Antimony	52 Te 127.6 Tellurium	53 I 126.9 Iodine	54 Xe 131.3 Xenon			
55 Cs 132.9 Caesium	56 Ba 137.3 Barium	Lanthanoids 89–103 Actinoids		73 Ta 180.9 Tantalum	74 W 183.9 Tungsten	75 Re 186.2 Rhenium	76 Os 190.2 Osmium	77 Ir 192.2 Iridium	78 Pt 195.1 Platinum	79 Au 197.0 Gold	80 Hg 200.6 Mercury	81 Tl 204.4 Thallium	82 Pb 207.2 Lead	83 Bi 209.0 Bismuth	84 Po Astatine	85 At	86 Rn			
87 Fr	88 Ra			105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Uut	114 Fl	115 Uup	116 Lv	117 Uus	118 Uuo			
Francium	Radium	Actinoids		Rutherfordium	Seaborgium	Bohrium	Hassium	Meitnerium	Darmstadtium	Roentgenium	Copernicium	Ununtrium	Flerovium	Ununpentium	Livermorium	Ununseptium	Ununoctium			

Lanthanoids

57 La 138.9 Lanthanum	58 Ce 140.1 Cerium	59 Pr 140.9 Praseodymium	60 Nd 144.2 Neodymium	61 Pm Promethium	62 Sm 150.4 Samarium	63 Eu 152.0 Europium	64 Gd 157.3 Gadolinium	65 Tb 158.9 Terbium	66 Dy 162.5 Dysprosium	67 Ho 164.9 Holmium	68 Er 167.3 Erbium	69 Tm 168.9 Thulium	70 Yb 173.1 Ytterbium	71 Lu 175.0 Lutetium
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Actinoids

89 Ac	90 Th 232.0 Thorium	91 Pa 231.0 Protactinium	92 U 238.0 Uranium	93 Np Neptunium	94 Pu Plutonium	95 Am Americium	96 Cm Curium	97 Bk Berkelium	98 Cf Californium	99 Es Einsteinium	100 Fm Fermium	101 Md Mendelevium	102 No Nobelium	103 Lr Lawrencium
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Standard atomic weights are abridged to four significant figures.
Elements with no reported values in the table have no stable nuclides.

Information on elements with atomic numbers 113 and above is sourced from the International Union of Pure and Applied Chemistry Periodic Table of the Elements (January 2016 version).
The International Union of Pure and Applied Chemistry Periodic Table of the Elements (February 2010 version) is the principal source of all other data. Some data may have been modified.

2016 HSC Physics Marking Guidelines

Section I, Part A

Multiple-choice Answer Key

Question	Answer
1	D
2	C
3	C
4	A
5	D
6	D
7	B
8	D
9	B
10	C
11	B
12	B
13	C
14	B
15	A
16	A
17	C
18	C
19	A
20	B

Section I, Part B

Question 21 (a)

Criteria	Marks
• Relates the reason for orbital decay to altitude of orbit	2
• Provides some relevant information	1

Sample answer:

Low-Earth orbit satellites encounter atmospheric drag due to their lower altitude. This drag reduces the orbital velocity causing orbital decay.

Question 21 (b)

Criteria	Marks
• Applies correct method to calculate magnitude and provides correct units	3
• Shows correct substitution into the equation without correct unit conversion OR without correct units	2
• Substitutes into a relevant formula	1

Sample answer:

$$F = \frac{Gm_1m_2}{r^2} = \frac{6.67 \times 10^{-11} \times 6 \times 10^{24} \times 50}{(8000 \times 10^3)^2} = 312.7N$$

Question 22 (a)

Criteria	Marks
• Identifies any motion of the magnet which would induce an oscillating voltage in coil B	2
• Identifies any motion of the magnet which would induce a voltage in coil B	1

Sample answer:

One end of the bar magnet is moved quickly and repeatedly towards and away from the end of coil B.

Question 22 (b)

Criteria	Marks
• Fully accounts for the features of the graph with reference to the motion of the magnet and the forces acting on it	4
• Relates features of the graph to the motion of the magnet	3
• Links a feature of the graph to a motion of the magnet	2
• Provides relevant information	1

Sample answer:

The magnet falls from rest and accelerates at 9.8 ms^{-2} due to gravity (downward line on graph). As the magnet approaches the disc, eddy currents are induced in the disc due to changes in magnetic flux. A magnetic field due to these currents opposes the change in flux, slowing the descent of the magnet (upward line on graph). It eventually comes to rest on the disc ($v = 0$).

Question 23 (a)

Criteria	Marks
• Correctly explains the role of deflection plates in showing different voltages	3
• Shows some understanding of how deflection plates work	2
• Provides some relevant information	1

Sample answer:

When an electron moves between the deflection plates it experiences a force due to the electric field. An increase in the input voltage increases the voltage across the deflection plates, increasing the electric field and causing more deflection that can be seen on the fluorescent screen.

Question 23 (b)

Criteria	Marks
Provides correct substitution for the calculation of B with the correct direction	3
- Provides correct substitution to calculate B OR - Equates $F = qE$ and $F = qvB$ and states correct direction	2
- Provides correct direction OR - Provides partial substitution into a relevant equation	1

Sample answer:

$$F_E = F_m$$

$$qE = qvB$$

$$B = \frac{E}{v}$$

$$B = \frac{10}{5.2 \times 10^4}$$

$$B = 0.0002T \text{ up the page}$$

Question 24

Criteria	Marks
Relates the graph to the behaviour of electrons both above and below the critical temperature	4
Relates the graph to the behaviour of electrons above or below the critical temperature	3
Links a feature of the graph to the behaviour of electrons	2
Provides some relevant information	1

Sample answer:

Above T_c , the wire is a normal conductor with electrical resistance due to collisions between electrons and the lattice. Since the lattice vibrates more at higher temperatures, these collisions increase with temperature and lead to higher electrical resistance at higher temperatures.

Below T_c , the wire becomes superconductive. Electrons form Cooper pairs and pass through the lattice without loss of energy, resulting in zero electrical resistance for all temperatures below T_c .

Question 25 (a)

Criteria	Marks
• Provides valid comparison between force and distance in the graphs	2
• Identifies a relationship between force and distance in one of the graphs	1

Sample answer:

Both graphs show that as distance increases, the force decreases. However, in Team A's graph, the force between the masses decreases at a decreasing rate, whereas in Team B's graph, the force decreases at a constant rate.

Question 25 (b)

Criteria	Marks
• Makes an informed judgement of the appropriateness of each data set	3
• Identifies strengths and/or weaknesses of the data set(s)	2
• Provides some relevant information	1

Sample answer:

Team A's data set has a good range but too few measurements for a valid relationship to be deduced.

Team B's data set has sufficient measurements but over an insufficient range of distances for a valid relationship to be deduced.

Question 26

Criteria	Marks
• Identifies applications and explains how they have affected society	5
• Identifies applications and outlines their effects on society	4
• Identifies applications and outlines an effect on society	3
• Identifies applications OR • Outlines an effect on society	2
• Provides some relevant information	1

Sample answer:

Transformers allow the widespread use of electronic devices powered by mains voltages. Such proliferation of electronic devices has had significant effects on society, including rapid communication and data processing resulting in significant changes to the workforce, and the development of a large IT sector.

Transformers allow the efficient distribution of electrical energy over distances, so that consumers may be located at a distance from the point of generation. This reduces the effect of pollution in large cities, improving living conditions and the focus of health programs. It also allows an equitable standard of living to be delivered across the population regardless of location.

Question 27 (a)

Criteria	Marks
• Outlines observations including equipment used	3
• Outlines an observation OR equipment used	2
• Provides some relevant information	1

Sample answer:

Hertz observed that if a glass screen between the spark emitter and the receiving loop antenna was removed, the maximum length of the spark across the air gap of the loop antenna was increased.

Answers could include:

The air gap of the loop antenna was set at the maximum for which a spark across it was maintained. Inserting a glass screen caused the spark to stop.

A diagram is provided showing a spark source and loop antenna with a spark gap with a glass screen between them.

Question 27 (b)

Criteria	Marks
• Shows a thorough understanding of how the investigation of the photoelectric effect changed the understanding of light, including one specific piece of evidence	4
• Relates the investigation of photoelectric effect to the understanding of light	3
• Shows some understanding of the investigation of the photoelectric effect and/or the change in understanding of light	2
• Provides some relevant information	1

Sample answer:

Before the explanation of the photoelectric effect, light was considered to consist of transverse waves of different wavelengths. One experimental result showed that the velocity of photoelectrons did not increase with increased light intensity, but did increase when the frequency of the light was increased.

This was not consistent with predictions of the wave model and could only be explained by considering the light to consist of discrete packets of energy (quanta) where quanta of high frequency light possessed more energy than those of low frequency light.

Question 28

Criteria	Marks
• Explains the change in acceleration of the rocket with reference to the law of conservation of momentum	5
• Links the changes in acceleration of the rocket to the law of conservation of momentum	4
• Shows some understanding of the law of conservation of momentum and/or the acceleration of the rocket	2–3
• Provides some relevant information	1

Sample answer:

The rocket is propelled forwards by expelling gases backwards.

The mutual force acting between the rocket and gases exerts an equal and opposite effect on each. Since this force is applied over the same time interval, the rocket and gases experience equal and opposite changes in momentum, that is, momentum is conserved.

Since the mass of the rocket is decreasing while the force acting on it due to the gases is constant, the acceleration of the rocket increases as it ascends.

Answers could include:

Algebraic analysis.

Question 29

Criteria	Marks
Provides explanations with reference to electric generators and TWO other technologies	6
Provides explanations with reference to electric generators and ONE other technology	5
Links relevant discoveries to electric generators and TWO other technologies	4
Links relevant discoveries to TWO technologies	3
- Identifies relevant discoveries OR - Outlines one relevant discovery	2
Identifies a discovery or a technology other than electric generators	1

Sample answer:

Faraday discovered that a change in magnetic flux would induce an emf in a conductor. This discovery led directly to the development of the electric generator.

Physicists discovered that electrons could be made to travel through evacuated tubes. They were deflected by electric and magnetic fields and could cause a fluorescent screen to emit light. The electrons could be manipulated precisely to form an image on the screen leading to the development of television.

Physicists made discoveries about the properties of semiconductors. They exhibited a small energy gap between conduction and valence bands that could be altered by doping. Doped semiconductor combinations (eg PNP) formed devices that behaved in the same way as existing thermionic components, allowing the miniaturisation of computers, and the development of practical computing.

Question 30 (a)

Criteria	Marks
• Applies a correct method to calculate the maximum possible energy released	3
• Provides the main steps	2
• Substitutes into a relevant formula	1

Sample answer:

$$\begin{aligned}
 \Delta E &= E_{p\text{final}} - E_{p\text{initial}} \\
 &= \left(\frac{-Gm_1m_2}{r_{\text{final}}} \right) - \left(\frac{-Gm_1m_2}{r_{\text{initial}}} \right) \\
 &= \frac{-6.67 \times 10^{-11} \times 6.39 \times 10^{23} \times 2}{3376203} - \frac{(-6.67 \times 10^{-11} \times 6.39 \times 10^{23} \times 2)}{3376204} \\
 &= -7.48 \text{ J (lost by falling mass)} \\
 \therefore \text{light bulb released a maximum of 7.48 J of energy}
 \end{aligned}$$

Answers could include:

Assuming g_{mars} is constant over this range,

$$\text{Work} = Fs$$

$$\begin{aligned}
 &= \frac{Gm_1m_2}{r^2} \times s \\
 &= \frac{-6.67 \times 10^{-11} \times 6.39 \times 10^{23} \times 2 \times 1}{3376203^2} \\
 &= 7.48 \text{ J}
 \end{aligned}$$

Question 30 (b)

Criteria	Marks
• Explains the difference in behaviour	3
• Provides some reasoning for the change in behaviour	2
• Identifies a change in behaviour	1

Sample answer:

When the switch is open, there is no longer any magnetic effect opposing the falling of the mass, so the mass now falls more quickly.

Answers may include:

Information on Lenz's Law or conservation of energy.

Section II

Question 31 (a) (i)

Criteria	Marks
• Relates what geophysicists do to the seismic data generated by a nuclear explosion	2
• Makes a relevant statement about seismic waves of nuclear origin	1

Sample answer:

Nuclear weapon explosions create a unique seismic signature, which is detected by seismographs. Geophysicists who analyse the seismograms can deduce whether and where a nuclear explosion has occurred and thus can monitor compliance with test ban treaties.

Question 31 (a) (ii)

Criteria	Marks
• Outlines one geophysical method and links an economic benefit to the method	3
• Identifies a geophysical method and a general economic benefit OR • Outlines a geophysical method	2
• Identifies a geophysical method	1

Sample answer:

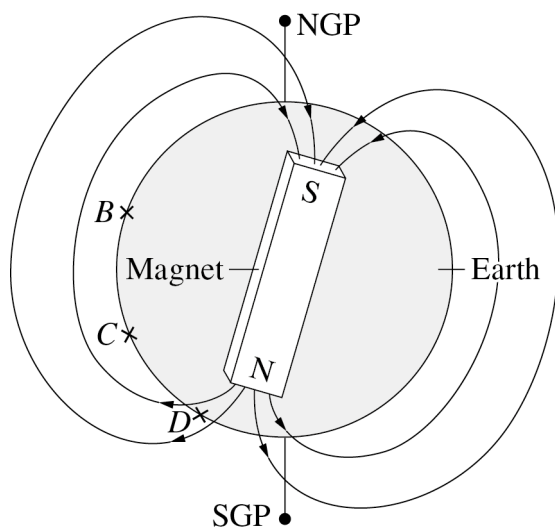
Oil deposits produce gravitational anomalies, which can be measured and used to locate the deposits. Oil extracted from these deposits provides energy for a great range of industries, such as transport. Hence this geophysical method has resulted in great economic benefits.

Question 31 (b) (i)

Criteria	Marks
• Describes a relevant model	3
• Outlines a relevant model	2
• Provides some relevant information	1

Sample answer:

One model that has been used for Earth's magnetic field postulates the existence of a permanent dipole, represented as:



At the equator (B), this model shows Earth's magnetic field being nearly parallel to Earth's surface. This model shows that as the latitude increases, $B \rightarrow C \rightarrow D$, the inclination of the magnetic field to Earth's surface increases.

Question 31 (b) (ii)

Criteria	Marks
• Describes features of Earth's magnetic field	3
• Outlines features of Earth's magnetic field	2
• Identifies a feature of Earth's magnetic field	1

Sample answer:

The Earth's magnetic field changes with time, which is observed in the change of position of the magnetic N/S poles over geologically short times.

Earth's magnetic field is not uniformly symmetrical, since it has some irregularities that are intrinsic due to local variations in composition and density and other irregularities that are extrinsic and change due to effects such as solar wind.

Question 31 (c)

Criteria	Marks
• Explains TWO gravimetric corrections	4
• Identifies TWO gravimetric corrections and gives a reason why a specific gravimetric correction must be made	3
• Identifies TWO gravimetric corrections OR • Gives a reason why a specific gravimetric correction must be made	2
• Identifies a feature of gravimetric analysis	1

Sample answer:

Analysis of gravimetric data must take the altitude at which the measurement was taken into account since the force of gravity decreases with altitude. Correction for this variable is called the free-air correction.

Corrections must also be applied for local variations in rock density and distribution, since these affect the gravimeter data. This is called the Bouguer correction.

Question 31 (d)

Criteria	Marks
• Provides valid inferences with justifications	4
• Provides valid inferences and justifies one of them	3
• Provides valid inferences OR • Provides a valid inference with justification	2
• Provides some relevant information	1

Sample answer:

The same patterns of reversal occur over a shorter distance at location B compared with A, so the rate of seafloor spreading at B is slower than the rate at A. This is based on the assumption that the same extrinsic field changes affect cooling rocks at both locations.

The rate of spreading is the same on either side of the ridge at both locations. This is evidenced by the symmetric patterns on either side of the mid-ocean ridge.

Answers could include:

The non-uniform width of regions with normal/reverse magnetisation may be due to either variation of the seafloor spreading rate or due to variation of the lengths of time between reversals, or both. Further evidence is required to make a better inference.

Question 31 (e)

Criteria	Marks
Relates features on the seismogram to the behaviour of different types of earthquake waves and Earth's internal structure	6
Relates features on the seismogram to the behaviour of different types of earthquake waves and/or Earth's internal structure	5
Links features of the seismogram to waves and/or Earth's internal structure	3–4
- Identifies features on the seismogram OR - Links a feature on the seismogram to a type of wave or Earth's internal structure	2
Provides some relevant information	1

Sample answer:

Seismograms at *W* and *X* show two events at the stations separated by ~7 minutes and ~11 minutes respectively. The first event in each case is the arrival of the P-waves, followed by the S-wave.

P-waves are longitudinal waves that travel faster than S-waves which are transverse waves. Since *X* is further from the earthquake focus, the arrival time difference between P and S waves is greater at *X*.

S-waves cannot travel through liquids and so there can be no liquid layers extending as far as the line joining the focus and *X*.

The record at *Y* only shows the arrival of the P wave and so it can't be deduced from this record if there's a liquid component between the focus and *Y*, since we don't know if the S-wave was detected at that location.

Answers could include:

The P/S wave records of the seismograph at *X* have a smaller amplitude than the corresponding waves at *W*. This is due to the effect of the inverse square law and absorption of energy by material in the Earth.

Question 32 (a) (i)

Criteria	Marks
• Identifies differences between hard and soft X-rays	2
• Identifies a property of hard or soft X-rays	1

Sample answer:

Hard X-rays have a higher energy than soft X-rays. Hard X-rays have a greater ability to penetrate human body structures than soft X-rays.

Question 32 (a) (ii)

Criteria	Marks
• States reasons for the widespread use of X-rays to image limbs	3
• States a reason for the widespread use of X-rays	2
• Identifies a feature of X-ray imaging	1

Sample answer:

X-ray imaging provides high contrast between soft tissue and bone, at sufficient resolution to detect fractures. X-ray technology is cheaper than most alternatives and does not expose the patient to as much ionising radiation as CT. For these reasons X-ray is widely used to image human limbs.

Question 32 (b) (i)

Criteria	Marks
• Describes the piezoelectric effect and its application in ultrasound	3
• Describes the piezoelectric effect OR its application in ultrasound	2
• Identifies a feature of the piezoelectric effect	1

Sample answer:

The piezoelectric effect is the inter-conversion of mechanical and electrical energy in a crystal. When a potential difference is applied across such a crystal, its shape will alter. Conversely, applying a force to the crystal will result in a potential difference across it.

Since these variations can occur at high frequencies, the piezoelectric effect is ideal for creating and detecting ultrasound waves used for medical imaging.

Question 32 (b) (ii)

Criteria	Marks
• Describes what happens to the intensity at each boundary with the support of calculations	3
• Provides some relevant calculations and/or shows an understanding of what happens	2
• Provides some relevant information	1

Sample answer:

At the first boundary (gel/soft tissue), none of the ultrasound pulse is reflected, since both media have the same acoustic impedance.

At the second boundary (soft tissue/bone), some of the pulse will be reflected, and some will continue into the bone. Quantitatively:

$$\frac{I_r}{I_0} = \frac{(Z_2 - Z_1)^2}{(Z_2 + Z_1)^2}$$

$$\frac{I_r}{I_0} = \frac{(6.732 \times 10^6 - 1.632 \times 10^6)^2}{(6.732 \times 10^6 + 1.632 \times 10^6)^2} = 0.3718$$

So, approximately 37% of the pulse is reflected at the soft tissue/bone boundary and approximately 63% of the pulse passes into the bone.

Question 32 (c)

Criteria	Marks
• Contrasts the uses of CAT imaging and Doppler ultrasound imaging, providing an example of how each is used	4
• Outlines a use of each technology	3
• Outlines a use of either technology	2
• Provides some relevant information	1

Sample answer:

Doppler ultrasound is used to examine functional characteristics of the human body, in particular the motion of body structures and fluids. Specific uses of Doppler ultrasound include determining the motion of the heart and determining the motion of blood, particularly near valves or blockages.

In contrast, CAT imaging examines structural characteristics of the human body, at very high resolutions. CAT provides excellent contrast between hard and soft tissues, and also different forms of soft tissue. Specific uses of CAT imaging include the detection of abdominal tumours and the diagnosis of damaged soft tissue such as cartilage.

Question 32 (d)

Criteria	Marks
• Describes how PET scan technique is used in diagnosis with reference to the interaction of positrons and electrons	4
• Outlines how PET scan is used with reference to the interaction of positrons and electrons	3
• Identifies some features of the PET scan technique	2
• Identifies a feature of the PET scan technique	1

Sample answer:

PET imaging is useful in diagnosis because it allows for the use of specific radiopharmaceuticals which can target specific metabolic processes, such as high rates of metabolism associated with cancerous tissues.

During PET imaging, the patient is injected with a radioisotope that releases positrons. These positrons undergo annihilation with nearby electrons in the body, releasing gamma radiation. The patient is surrounded by a circular gamma detector that allows the distribution of the radioisotope within the patient to be computed.

Question 32 (e)

Criteria	Marks
• Explains how principles of physics are applied in MRI	6
• Relates principles of physics to their use in MRI	5
• Describes the use of a principle of physics used in MRI	4
• Describes a principle of physics used in MRI OR • Identifies principles of physics used	3
• Outlines features of MRI OR • Identifies a principle of physics used in MRI	2
• Provides some relevant information	1

Sample answer:

The physical principles related to spin and resonance are applied in MRI.

The magnetic properties of nucleons are related to their spin. Hydrogen is abundant in the human body and has a net spin that is determined by its single proton. During MRI a patient is placed in a strong magnetic field causing the spin directions of the hydrogen nuclei to align and precess about the magnetic field lines at a specific frequency.

Due to the principle of resonance, a radio wave at this frequency will be readily absorbed by the nuclei, causing their precession to become synchronised and some of them to be placed into a high energy state. This is necessary in order to detect the nuclei in subsequent processes.

Answers could include:

- The principle of electromagnetic induction in the detecting coils.
- The principle of superconductors in the field coil.
- The principles of semiconductor operation essential for the operation of computers used in MRI.

Question 33 (a) (i)

Criteria	Marks
Correctly distinguishes between resolution and sensitivity for optical telescopes	2
Shows some understanding of sensitivity OR resolution	1

Sample answer:

Resolution of an optical telescope is the ability to distinguish two close objects as separate images.

Sensitivity is the light-gathering power or the ability to detect faint objects for observation. This depends on the area of the mirror or lens of the telescope.

Question 33 (a) (ii)

Criteria	Marks
Outlines ways	3
- Identifies ways OR - Outlines one way	2
Provides some relevant information	1

Sample answer:

The resolution of ground-based telescopes can be improved by a number of techniques including active optics, adaptive optics and interferometry.

Active optics uses a slow computer-controlled feedback system to correct for sagging or deformities of large primary mirrors in optical telescopes.

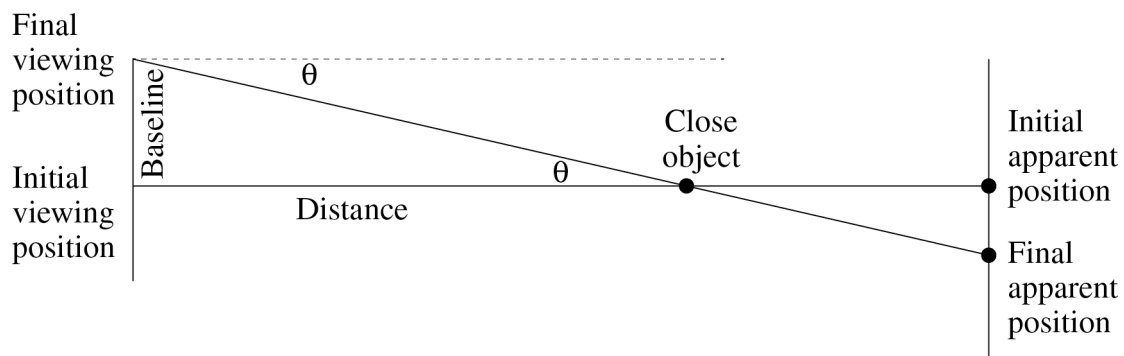
Interferometry uses data (optical or radio) from a number of collectors to produce an interference pattern which is then mathematically analysed to produce an 'unblurred' image.

Question 33 (b) (i)

Criteria	Marks
• Provides a correct explanation	3
• Shows some understanding of trigonometric parallax	2
• Provides some relevant information	1

Sample answer:

Trigonometry can be used to calculate the distance to relatively close stellar objects by using their apparent change in position with time, against background distant stars (parallax).



Viewing positions could change as the Earth rotates (diurnal) or as Earth orbits around the Sun (six months).

Distance can then be calculated by:

$$\text{Distance} = \text{Baseline} / \tan \theta$$

Question 33 (b) (ii)

Criteria	Marks
• Applies correct procedure to calculate the absolute magnitude	3
• Provides some main steps	2
• Substitutes into a relevant formula	1

Sample answer:

$$M = ? \quad m = 4.2 \quad p = 0.055 \quad d = ?$$

$$p = \frac{1}{d}$$

$$= \frac{1}{0.055}$$

$$= 18.18 \text{ parsecs}$$

$$M = m - 5 \log \left(\frac{d}{10} \right)$$

$$= 4.2 - 5 \log \left(\frac{18.18}{10} \right)$$

$$= 4.2 - 1.298$$

$$= 2.9$$

Absolute magnitude is 2.9

Question 33 (c)

Criteria	Marks
• Describes the use of spectral analysis to determine chemical composition and surface temperature of stars	4
• Outlines features of spectra related to composition and surface temperature	3
• Makes some reference to spectra to identify composition AND/OR temperature	2
• Provides some relevant information	1

Sample answer:

Particles in the atmosphere of stars absorb specific frequencies of light allowing the elements and molecules to be identified. Hence chemical composition can be determined.

The presence of molecules, neutral atoms or ionised atoms in a star are determined by its temperature. These can be detected in the absorption spectrum, allowing their temperature to be deduced.

Answers could include:

Information on Wien's Law.

Question 33 (d)

Criteria	Marks
• Explains how the study of different types of variable stars has contributed to the understanding of celestial objects	4
• Outlines features of different types of variable stars and relates a feature to the understanding of celestial objects	3
• Identifies features of variable stars OR • Relates a feature of variable stars to the understanding of a celestial object	2
• Provides some relevant information	1

Sample answer:

The period–luminosity relationship of Cepheid variables allows their distances from Earth to be deduced. Knowing this distance, inferences about celestial objects near the Cepheid can be made, such as absolute magnitude and relative age of star clusters.

Features of binary and multiple star systems, such as orbital period, mass and relative size, can be deduced from their extrinsic variations in brightness.

Answers could include:

Use of variables to identify exoplanets.

Question 33 (e)

Criteria	Marks
• Relates processes that occur during stellar evolution to the non-uniform distribution of stars on the H–R diagram	6
• Relates processes that occur during stellar evolution to the distribution of stars on the H–R diagram	5
• Relates processes that occur in stars to their position on the H–R diagram	4
• Describes processes that occur in stars OR • Relates a process that occurs in stars to their position on the H–R diagram OR • Describes characteristics of stars in the main regions on the H–R diagram	3
• Describes a process that occurs in stars OR • Identifies regions on the H–R diagram	2
• Provides some relevant information	1

Sample answer:

Stars on an H–R diagram are non-uniformly distributed into distinct groups dependent on processes which occur during their evolution.

The primary source of energy in stars on the main sequence is hydrogen fusion.

In the next evolutionary stage for most stars, helium fusion is the main source of energy. These belong to the group known as red giants.

The next evolutionary stage for most stars is characterised by gravitational collapse during which gravitational potential energy is converted to radiant energy. These belong to the group known as white dwarfs.

Since the transition from one process to another occurs quickly relative to time spent in each group, fewer stars are distributed in areas outside of these groups on the H–R diagram.

Answers could include:

- Reference to other groups such as protostars, supergiants
- Sketch of H–R diagram
- Reference to properties of stars related to their distribution within particular groups
- Globular and open clusters.

Question 34 (a) (i)

Criteria	Marks
• Identifies TWO nucleons and states a difference	2
• Provides some relevant information	1

Sample answer:

The two types of nucleons are neutrons and protons. Protons are positively charged whereas neutrons have no charge.

Question 34 (a) (ii)

Criteria	Marks
• Explains the stability in terms of two forces	3
• Explains the stability due to one force OR • Links stability to two forces	2
• Provides some relevant information	1

Sample answer:

Protons carry a positive charge so that there is electrostatic repulsion that occurs between the protons in the nucleus.

The strong nuclear force is needed to hold the nucleus together. It occurs between proton–proton, neutron–neutron and protons–neutrons. It is a short range force that is strongly attractive when the nucleons are 0.5 to 1.0 the diameter of a proton apart; anything outside this range will be repulsive.

Question 34 (b) (i)

Criteria	Marks
• Outlines features	3
• Outlines one feature or identifies features	2
• Provides some relevant information	1

Sample answer:

Bohr's model cannot explain why some spectral lines were observed to be more intense than others; it does not explain why some transitions were favoured over others.

With more sensitive instruments, other lines were observed known as hyperfine lines. This indicates there is some splitting of the energy level that Bohr's model cannot explain.

Question 34 (b) (ii)

Criteria	Marks
• Applies correct procedure to calculate n_i	3
• Provides some main steps	2
• Substitutes into a relevant formula	1

Sample answer:

$$\frac{E}{hc} = \frac{1}{\lambda} = R \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right) \text{ therefore, } \frac{1}{n_i^2} = \frac{1}{n_f^2} - \frac{E}{hcR}$$

$$\frac{1}{n_i^2} = \frac{1}{2^2} - \frac{4.089 \times 10^{-19}}{6.626 \times 10^{-34} \times 3 \times 10^8 \times 1.097 \times 10^7} \text{ so } \frac{1}{n_i^2} = 0.06248 \text{ therefore,}$$

$$n_i = \sqrt{\frac{1}{0.06248}} = 4$$

Question 34 (c)

Criteria	Marks
• Describes both Pauli's and Heisenberg's ideas and how they changed the Bohr model	4
• Describes both Pauli's and Heisenberg's ideas and outlines a change made by either Pauli or Heisenberg	3
• Describes either Pauli's or Heisenberg's ideas OR • Outlines a change to Bohr's model	2
• Provides some relevant information about Pauli's or Heisenberg's ideas	1

Sample answer:

Pauli introduced the last quantum number associated with spin. His exclusion principle states that no two electrons can have the same quantum number. This changed Bohr's model by limiting the number of electrons that can exist in each shell.

Heisenberg's uncertainty principle states that if you know the accurate position of the particle then you cannot know its momentum accurately. If you can determine its momentum accurately then you cannot specify its position accurately. This changed Bohr's shell to an orbital probability cloud since precise measurements cannot be made.

Question 34 (d)

Criteria	Marks
• Relates evidence from accelerators and detectors to development of the standard model	4
• Describes the purpose of accelerators and detectors in identifying subatomic particles or the development of the standard model	3
• Describes the purpose of accelerators and detectors OR • Describes the standard model	2
• Provides some relevant information	1

Sample answer:

High energy collisions in particle accelerators produced many different types of subatomic particles which had not been previously observed. Properties of these particles, such as momentum and charge, could be deduced from measurements made using a range of detectors.

The standard model was developed to provide a framework for understanding these new particles. It also predicted the existence of particles which were subsequently produced and detected, providing further evidence to support the standard model.

Question 34 (e)

Criteria	Marks
• Relates Fermi's work to applications of nuclear physics and their effects on society	6
• Relates the work of Fermi to application(s) and effect(s) on society	5
• Relates the work of Fermi to any application and its effect on society	4
• Describes the work of Fermi, identifies an application and an effect on society	3
• Identifies an application and its effect on society OR • Identifies applications OR • Describes the work of Fermi	2
• Provides some relevant information	1

Sample answer:

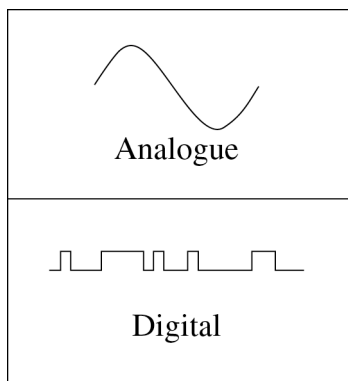
Fermi made experimental observations of nuclear reactions in the first nuclear fission reactor. He was able to determine the requirements for a controlled chain reaction, leading to the development of stable nuclear reactors.

Nuclear reactors are used for the production of electricity using fission. This provides an alternative reliable supply of energy for a significant number of people in many countries.

Nuclear reactors are also used to produce radioisotopes. These may be used in medical imaging and in the treatment of cancer, enhancing healthcare.

Question 35 (a) (i)

Criteria	Marks
• Shows voltage outputs of both analogue and digital signals	2
• Identifies an analogue or digital signal	1

Sample answer:**Question 35 (a) (ii)**

Criteria	Marks
• Describes an advantage of electric circuits and an advantage of electronic circuits	3
• Describes an advantage of either electric or electronic circuits	2
• Identifies an advantage of electronic circuits OR electric circuits	1

Sample answer:

Electronic circuits can work with tiny voltages and currents and can be miniaturised to allow large amounts of processing in a small volume.

Electric circuits can carry large amounts of electrical energy, making them ideal for transfer of energy to power various devices. Electric circuits are much more simple and cannot process signals in the same way as electronic circuits.

Question 35 (b) (i)

Criteria	Marks
• Compares all features	3
• Compares some features	2
• Provides some relevant information	1

Sample answer:

This op amp does not have ideal properties. An ideal amplifier has infinite open loop gain and infinite input impedance, while this op amp has a very high open loop gain.

The output impedance of an ideal amplifier is zero, but this op amp has a relatively low but not insignificant output resistance.

An ideal amplifier has an infinite bandwidth, but this amplifier has a limited bandwidth.

Question 35 (b) (ii)

Criteria	Marks
• Draws correct circuit diagram with correct feedback resistor	3
• Draws a relevant circuit and/or shows understanding to calculate the value of feedback resistor	2
• Provides some relevant information	1

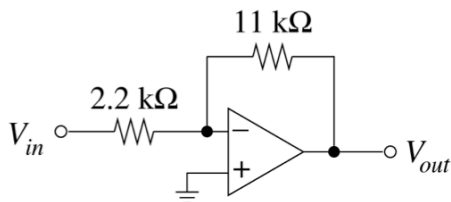
Sample answer:

$$\frac{V_{out}}{V_{in}} = \frac{R_f}{R_i}$$

$$\frac{-10}{2} = \frac{-R_f}{2.2}$$

$$R_f = 2.2 \times \frac{10}{2}$$

$$R_f = 11\text{k}\Omega$$



Question 35 (c)

Criteria	Marks
Completes a relevant truth table and determines correct voltages for each input	4
- Provides correct truth table OR - Determines input voltages using an incorrect truth table	3
Partially completes the truth table	2
Shows a basic understanding of truth tables	1

Sample answer:

<i>A</i>	<i>B</i>	<i>C</i>	<i>Output</i>
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	1

The output would be low only when inputs *A* and *B* are between 2U–5V and input *C* is less than 0.8 V.

Answers could include:

A low voltage can be any value $0 \leq V \leq 0.8$

A high voltage can be any value $2 \leq V \leq 5$

Question 35 (d)

Criteria	Marks
• Explains advantages of using integrated circuits over individual transistors in electronic devices	4
• Outlines the advantages OR explains an advantage of integrated circuits over individual transistors	3
• Identifies features of integrated circuits and/or individual transistors	2
• Provides some relevant information	1

Sample answer:

Many individual transistors are needed to perform even simple calculations. Therefore, early computers using transistors were bulky, less powerful and less reliable than their integrated circuit counterparts. The integrated circuit consists of many tiny transistors all built onto the one silicon chip. This allows modern computers to be more compact, efficient and cost-effective when compared with previous technology based on individual transistors.

Question 35 (e)

Criteria	Marks
• Explains THREE changes	6
• Identifies THREE changes that need to be made and explains at least two	5
• Outlines changes that need to be made	4
• Identifies changes that need to be made and outlines at least one	3
• Identifies changes that need to be made OR • Outlines a change that needs to be made	2
• Provides some relevant information	1

Sample answer:

Use a Relay – The fan is stated as 240 V. This would require a current and voltage greater than what can be supplied by an electronic circuit. In this case, a relay should be used in place of the fan and the fan can be driven by a separate circuit through the relay.

R_1 should be 5 k Ω – Op amp in open loop configuration will act as comparator. It will switch between high and low when the input voltage equals the reference voltage. In this case, the reference voltage is 7.5 V due to R_2 and R_3 being equal. However the input voltage will never match this as R_1 is too great. To match 7.5 V, R_1 should be equal to the required thermistor value. From the graph the thermistor will be 5 k Ω at 30° so R_1 should also be 5 k Ω .

Swap the positions of R_1 and the thermistor. Since the resistance decreases with temperature, an increase in temp will tend to pull down the input voltage. This means the fan will be on below 30° and switch off above 30° which is the opposite of what is required. Swapping R_1 and the thermistor would rectify this.

Answers could include:

- Adjusting the reference voltage by changing the values of either R_2 and R_3 to suit the R_1 / thermistor voltage divider
- Swapping the two inputs of the op amp.

2016 HSC Physics Mapping Grid

Section I Part A

Question	Marks	Content	Syllabus outcomes
1	1	9.3.4.2.1	H7, H9
2	1	9.3.3.3.3	H4, H7
3	1	9.4.1.2.3, 9.4.1.3.3	H9
4	1	9.4.4.3.1	H3
5	1	9.4.1.3.3	H9
6	1	9.2.4.2.6	H10
7	1	9.3.2.2.7	H7, H9
8	1	9.3.2.3.1/2, 9.3.2.2.3/4	H7, H9
9	1	9.3.2.2.5/6	H7, H9
10	1	9.2.4.2.9	H6
11	1	9.4.2.2.6, 9.4.2.3.4	H10
12	1	9.4.1.3.2, 9.4.1.2.3	H9
13	1	9.4.2.2.6	H10
14	1	9.2.2.3.5	H6
15	1	9.4.3.3.4	H7
16	1	9.3.2.2.3/4, 9.4.1.2.5	H7, H9
17	1	9.2.2.2.1, 9.2.4.3.2	H6, H9
18	1	9.2.1.3.3, 9.2.2.2.8	H6
19	1	9.2.4.2.9	H6
20	1	9.3.1.2.3, 9.3.1.3.3	H7, H9

Section I Part B

Question	Marks	Content	Syllabus outcomes
21 (a)	2	9.2.2.2.11	H9
21 (b)	3	9.2.3.3.2	H9
22 (a)	2	9.3.2.3.2	H9
22 (b)	4	9.3.2.3.4, 9.2.1.3.3	H9
23 (a)	3	9.4.1.2.9	H9
23 (b)	3	9.4.1.3.3	H9
24	4	9.4.4.2.5/4/3	H7
25 (a)	2	9.2.3.2.2, 9.2.3.3.2, H14.1a	H6, H14
25 (b)	3	9.2.3.2.2, 9.2.3.3.2, H14.1c	H6, H14
26	5	9.3.4.2.1/7	H4
27 (a)	3	9.4.2.2.1	H1

Question	Marks	Content	Syllabus outcomes
27 (b)	4	9.4.2.1.1, 9.4.2.3.2	H2
28	5	9.2.2.2.7	H6
29	6	9.2, 9.3, 9.4	H3
30 (a)	3	9.2.1.2.3	H6
30 (b)	3	9.3.2.2.5	H7

Section II

Question	Marks	Content	Syllabus outcomes
Question 31		Geophysics	
31 (a) (i)	2	9.5.5.2.2	H4
31 (a) (ii)	3	9.5.5.2.1	H3
31 (b) (i)	3	9.5.4.3.1, 9.5.4.3	H9
31 (b) (ii)	3	9.5.4.2.1/2	H9
31 (c)	4	9.5.2.2.8	H1
31 (d)	4	9.5.4.3.2	H14
31 (e)	6	9.5.3.3.3	H12
Question 32		Medical Physics	
32 (a) (i)	2	9.6.2.2.2	H8
32 (a) (ii)	3	9.6.4.3.4	H4
32 (b) (i)	3	9.6.1.2.2	H7
32 (b) (ii)	3	9.6.1.2.6	H13
32 (c)	4	9.6.4.3.4, 9.6.1.3.3	H3
32 (d)	4	9.6.3.2.3/4/5	H3
32 (e)	6	9.6.4	H3
Question 33		Astrophysics	
33 (a) (i)	2	9.7.1.2.3	H8
33 (a) (ii)	3	9.7.1.2.5	H3
33 (b) (i)	3	9.7.2.2.2	H3
33 (b) (ii)	3	9.7.4.3.1	H12
33 (c)	4	9.7.3.2.5	H8
33 (d)	4	9.7.5.2.3, 9.7.5.2.4	H4
33 (e)	6	9.7.6.3.1/2/3	H13
Question 34		From Quanta to Quarks	
34 (a) (i)	2	9.8.3.2.1	H2
34 (a) (ii)	3	9.8.3.2.7/8	H9
34 (b) (i)	3	9.8.1.3.4	H10
34 (b) (ii)	3	9.8.1.3.3	H7, H12
34 (c)	4	9.8.2.3.1	H1
34 (d)	4	9.8.4.2.4/5	H2
34 (e)	6	9.8.3, 9.8.4, 9.8.3.2.5/9/10, 9.8.4.2.2, 9.8.4.3.1	H3, H4

Question	Marks	Content	Syllabus outcomes
Question 35		The Age of Silicon	
35 (a) (i)	2	9.9.2.3.1	H8
35 (a) (ii)	3	9.9.2.2.1	H7
35 (b) (i)	3	9.9.6.2.1, 9.9.6.2.4	H2
35 (b) (ii)	3	9.9.6.2.8, 9.9.6.2.7, 9.9.6.3.3	H7
35 (c)	4	9.9.5.2.1, 9.9.3.1, 9.9.5.3.2	H7
35 (d)	4	9.9.1.2.2	H3
35 (e)	6	9.9.4.2.1, 9.9.3.2.5, 9.9.3.3.2, 9.9.2.2.5, 9.9.6.2.6	H7